

MultiLabelMarginLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.MultiLabelMarginLoss.html>

Description:

Creates a criterion that optimizes a multi-class multi-classification hinge loss (margin-based loss) between input `xxx` (a 2D mini-batch Tensor) and output `yyy` (which is a 2D Tensor of target class indices). For each sample in the mini-batch: where $x \in \{0, \dots, x.size(0)-1\}$, $y \in \{0, \dots, y.size(0)-1\}$, $0 \leq y[i] \leq x.size(0)-1$, and $i \neq y[j] \Rightarrow y[i] \neq y[j]$ for all i and j . `yyy` and `xxx` must have the same size. The criterion only considers a contiguous block of non-negative targets that starts at the front. This allows for different samples to have variable amounts of target classes.

Function Signature

```
class torch.nn.MultiLabelMarginLoss(size_average=None,  
reduce=None, reduction='mean')[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> loss = nn.MultiLabelMarginLoss()  
>>> x = torch.FloatTensor([[0.1, 0.2, 0.4, 0.8]])  
>>> # for target y, only consider labels 3 and 0, not after  
label -1  
>>> y = torch.LongTensor([[3, 0, -1, 1]])  
>>> # 0.25 * ((1-(0.1-0.2)) + (1-(0.1-0.4)) + (1-(0.8-0.2)) +  
(1-(0.8-0.4)))  
>>> loss(x, y)  
tensor(0.85...)
```

Detailed Documentation

Documentation

Creates a criterion that optimizes a multi-class multi-classification hinge loss (margin-based loss) between input `xxx` (a 2D mini-batch Tensor) and output `yyy` (which is a 2D Tensor of target class indices). For each sample in the mini-batch:

where $x \in \{0, \dots, x.size(0)-1\}$, $x \in \{0, \dots, \text{x.size}(0) - 1\}$, $y \in \{0, \dots, y.size(0)-1\}$, $y \in \{0, \dots, \text{y.size}(0) - 1\}$, $0 \leq y[j] \leq x.size(0)-1$, $0 \leq y[j] \leq \text{x.size}(0)-1$, and $i \neq y[j]$, $i \neq y[j]$ for all i and j .

`yyy` and `xxx` must have the same size.

The criterion only considers a contiguous block of non-negative targets that starts at the front.

This allows for different samples to have variable amounts of target classes.

`size_average` (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to `False`, the losses are instead summed for each minibatch. Ignored when `reduce` is `False`. Default: `True`

`reduce` (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on `size_average`. When `reduce` is `False`, returns a loss per batch element instead and ignores `size_average`. Default: `True`

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: (C)(C)(C) or (N,C)(N, C)(N,C) where N is the batch size and C is the number of classes.

Target: (C)(C)(C) or (N,C)(N, C)(N,C), label targets padded by -1 ensuring same shape as the input.

Output: scalar. If reduction is 'none', then (N)(N)(N).

Runs the forward pass.